MATTHEW LANDON

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ML and Software Engineer specializing in high-throughput automation pipelines, distributed data infrastructure, and research-grade predictive modeling. Tech expertise spans deep learning, multi-modal sensor fusion, network science, and full-stack API systems applied across fintech, aerospace, and planetary sciences.

EDUCATION

Rochester Institute of Technology — Rochester, NY
M.S. in Artificial Intelligence, GPA: 4.00 — *Expected May 2027*

Haverford College — Haverford, PA
B.S. in Applied Mathematics, GPA: 3.85 — *May 2025*

SKILLS

Languages & Tools: Python, Java, JavaScript, R, SQL, PySpark, FastAPI, Flask, Node.js, MATLAB, Git, LaTeX

ML/AI: PyTorch, TensorFlow, PySpark, Seaborn, Computer Vision (YOLOv8, OpenCV, ResNet), Deep Reinforcement Learning (Double DQN), Supervised/Unsupervised Learning, Reinforcement Learning, Natural Language Processing, Time-Series Forecasting, Network Graph Science, MC Dropout Epistemic Uncertainty

Cloud & Infrastructure: AWS (EC2, S3), Docker, Google Apps Script CLI (clasp), REST APIs, Linux, Jupyter, visNetwork

RELEVANT EXPERIENCE

AI Engineer Intern — Feb 2026 – Present
Nixacom AI — *Remote*

- **Fintech & Freight Quoting Infrastructure:** Architected a modular frontend/backend Request for Quote (RFQ) freight engine (nixacom-rfq-freight-quoting), writing production-grade data pipelines in Python and Java to automate core asset-verification workflows.
- **Computer Vision & Personalization:** Engineered end-to-end computer vision pipelines (YOLOv8, OpenCV) to segment and ingest unstructured media stream data, feeding downstream recipe-generation engines utilizing real-time object detection.
- **Procurement & Bid Automation:** Built scalable corporate procurement scraper scripts managed via the Google Apps Script Command Line Interface (clasp) (nixacom-rfq-test-bidding), processing corporate/government RFPs and compiling vendor proposals in standard TSU format.

EPIIC Research Intern — Jun 2024 – Present
Carnegie Science Earth and Planets Laboratory — *Washington, D.C.*

- **Network Graph Machine Learning:** Developed unsupervised network machine learning pipelines in R to map multidimensional cosine-similarity exoplanet vector graphs, discovering 5 distinct celestial subpopulations.
- **Statistical Error Control:** Leveraged high-dimensional regression and implemented the Benjamini-Hochberg False Discovery Rate (FDR) correction procedure to strictly isolate statistically valid stellar metallicity-radius correlations.
- **Observational Bias Quantification:** Modeled survey detection bias using Kruskal-Wallis variance testing and created interactive graph visualizations (visNetwork) to isolate physical anomalies from observational selection effects.

PUBLICATIONS & PRESENTATIONS

- **Data-Driven Characterization of Hot Jupiters via Network Analysis** — Carnegie Institution for Science, Under Review (Presented at AGU24 Annual Meeting, Feb 2026)
- **ANS: Uncertainty-Aware Autonomous Navigation for Mars Rovers During Communication Blackout** — Submitted to NeurIPS 2026, RIT, Apr 2026
  - Developed a 4-stage perception and planning architecture optimizing mission completion to 94% with a low 6% collision rate over 50 evaluation episodes. Formulated a real-time Hazard Score (H-score) cost map blending MC Dropout epistemic uncertainty (predictive entropy via 30 stochastic passes) from a MobileNetV3-Small classifier with a custom camera-LiDAR sensor fusion equation. Implemented a 3-layer navigation stack: Theta* any-angle global reference pathing, a Double DQN agent for reactive step-by-step driving (5k training episodes), and D* Lite overrides to mitigate perception conflicts.
- **Predicting Solar Flares: Standard ML vs. Transformer-Based ML Approaches** — RIT, Term Project, Dec 2025
  - Benchmarked Random Forest, XGBoost, KNN, and a custom Transformer-based CNN/LSTM (SolarFlareNet) on 120k training samples of NASA SDO/SHARP magnetogram time-series data profiles. Achieved operational-grade forecasting across all models (TSS > 0.50), with Random Forest reaching F1 = 0.906 (TSS = 0.801) for C-class flares and KNN achieving F1 = 0.883 (TSS = 0.745) for M-class flares. Validated real-time deployment readiness by verifying sub-100ms inference execution speed across all tested models.
- **Senior Thesis: Independence and Survivability of Space Habitats via ML Optimization and Differential Equations** — Haverford College, 2024–25

RESEARCH PROJECTS

- Establishing an Exoplanet Habitability Index via Supervised ML — built PCA/SVD-based models predicting Earth Similarity Index across 3,500+ planets (Haverford, 2024).